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Questions

Q1.

Which of these changes has the largest **increase** in the entropy of the system?

- ☒ **A** $H_2O(s) \rightarrow H_2O(l)$
- ☒ **B** $Hg(l) \rightarrow Hg(g)$
- ☒ **C** $H_2O(l) + HCl(g) \rightarrow H_3O^+(aq) + Cl^-(aq)$
- ☒ **D** C(graphite) → C(diamond)

(Total for question = 1 mark)

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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for an exothermic reaction is shown.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Which of these is true?

- ☒ **A** ΔH is positive
- ☒ **B** $\Delta S_{surroundings}$ is positive
- ☒ **C** ΔS_{system} is negative
- ☒ **D** ΔS_{total} is negative

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Q3.

This question is about some compounds of bromine.

(a) Potassium bromate(V) decomposes to form potassium bromide and oxygen.

$$2KBrO_3(s) \rightarrow 2KBr(s) + 3O_2(g) \qquad \Delta_r H^{\ominus} = -67.2\,kJ\,mol^{-1}$$

The standard molar entropies of these substances are given in the table.

Substance	$KBrO_3(s)$	$KBr(s)$	$O_2(g)$
$S^{\ominus}/JK^{-1}mol^{-1}$	149.2	95.9	205.0

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Calculate the total entropy change, ΔS_{total}, for this reaction at 298 K.

(5)

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(b) Bromide ions react with bromate(V) ions in acidic solution.

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

Two experiments are carried out.

(i) **Experiment 1**

The concentration of Br^- ions is determined at different times.

The concentrations of BrO_3^- ions and H^+ ions are in large excess and effectively constant.

The graph of concentration of Br^- ions against time is shown.

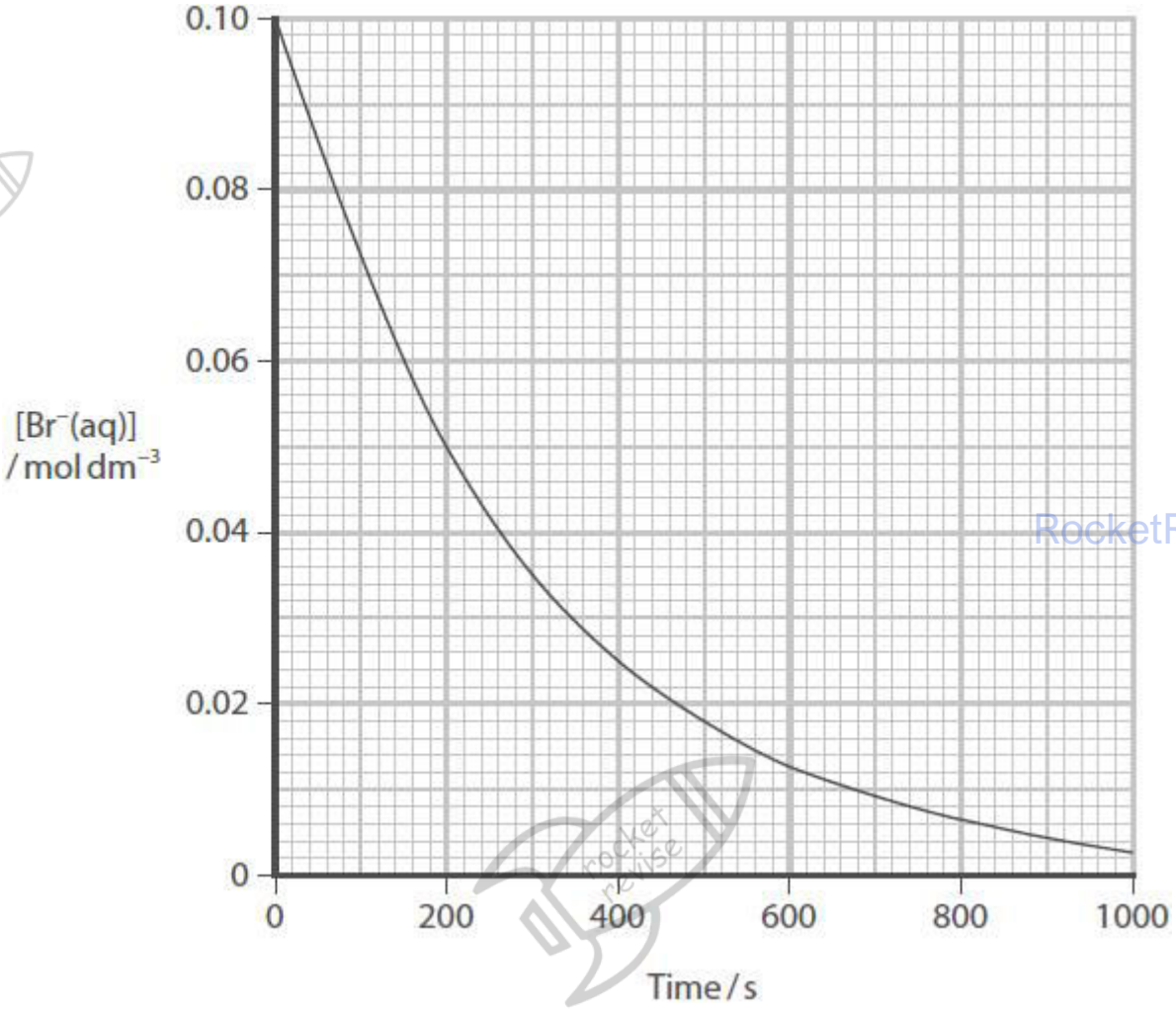
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Determine the order of the reaction with respect to bromide ions.
Show your working on the graph.

(3)

(ii) **Experiment 2**

The initial concentrations of BrO_3^- ions and H^+ ions are changed and the initial rate of reaction is determined.

The initial concentration of Br^- ions is constant and in large excess.

Run	$[BrO_3^-(aq)]$ / mol dm^{-3}	$[H^+(aq)]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.1	0.1	3.6×10^{-3}
2	0.2	0.1	7.2×10^{-3}
3	0.3	0.2	4.3×10^{-2}

Determine the order of reaction with respect to BrO_3^- ions and to H^+ ions.
You must explain your working.

(3)

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(iii) Give the overall rate equation for this reaction.
Include the units for the rate constant.

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(2)

(c) The rate constant for the reaction between bromoalkane and cyanide ions is determined at five different temperatures.

The results are given in the table.

Temperature (T) / K	1/Temperature ($1/T$) / K^{-1}	Rate constant (k) / s^{-1}	ln k
300	3.33×10^{-3}	3.72×10^{-5}	−10.20
310	3.23×10^{-3}	1.34×10^{-4}	−8.92
320	3.13×10^{-3}	5.48×10^{-4}	−7.51
330	3.03×10^{-3}	2.01×10^{-3}	−6.21
340	2.93×10^{-3}	7.23×10^{-3}	−4.93

Plot a graph of ln k against $1/T$ and use it to determine the activation energy, E_a.
Include the sign and units of the gradient and the activation energy.

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(7)

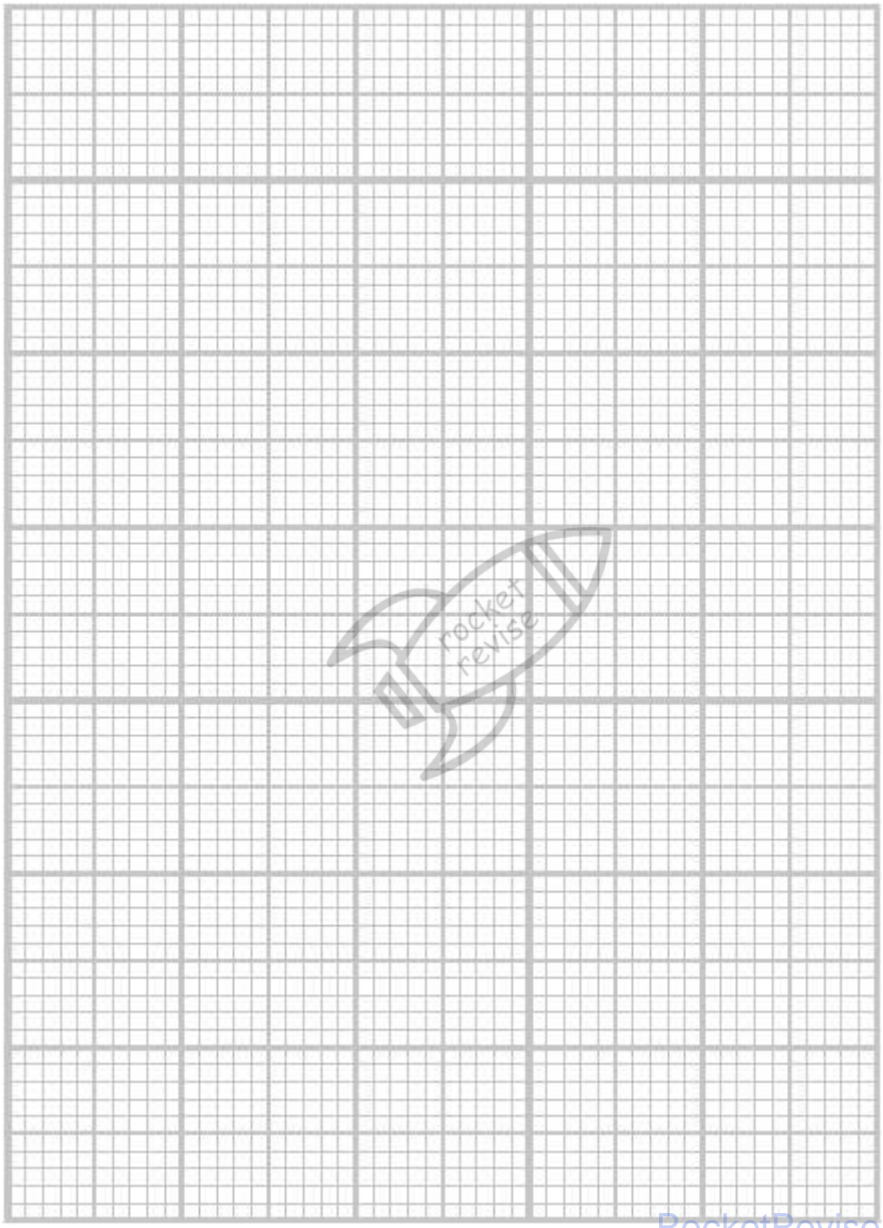
The Arrhenius equation can be expressed as

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant}$$

[R = 8.31 J mol^{-1} K^{-1}]

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(Total for question = 20 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The entropy change of the surroundings, $\Delta S_{surroundings}$, and the entropy change of the system, ΔS_{system}, for four different reactions are given.

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Reaction	$\Delta S_{surroundings}$ / J K^{-1} mol^{-1}	ΔS_{system} / J K^{-1} mol^{-1}
P	+245	+34
Q	+350	−276
R	−482	+65
S	−563	−128

Which of these is thermodynamically feasible?

☒ **A** reaction **P** only

☒ **B** reactions **P** and **Q** only

☒ **C** reaction **R** only

☒ **D** reactions **R** and **S** only

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(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The decomposition of calcium carbonate is an important reaction in the manufacture of cement.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \qquad \Delta H^{\ominus} = +178 \text{ kJ mol}^{-1}$$

What are the signs of the entropy change of the system, ΔS_{system}, and of the entropy change of the surroundings, $\Delta S_{surroundings}$?

	Sign of ΔS_{system}	Sign of $\Delta S_{surroundings}$
☒ **A**	positive	positive
☒ **B**	positive	negative
☒ **C**	negative	positive
☒ **D**	negative	negative

(Total for question = 1 mark)

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Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these gases would have the greatest standard molar entropy?

☒ **A** NH_3

☒ **B** H_2

☒ **C** N_2

☒ **D** SO_2

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(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ammonium nitrate is very soluble in water.

$$NH_4NO_3(s) + aq \rightarrow NH_4^+(aq) + NO_3^-(aq) \qquad \Delta H^{\ominus} = +25.8\ kJ\ mol^{-1}$$

What is the best explanation for this?

☒ **A** all ammonium salts are soluble in water

☒ **B** the activation energy of the reaction is very low

☒ **C** the enthalpies of hydration of the ions are very exothermic

☒ **D** the entropy change of the system, ΔS_{system}, is positive

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the standard entropy change of the system, in $J\ K^{-1}\ mol^{-1}$, for the reaction between nitrogen and hydrogen to form ammonia?

$$N_2 + 3H_2 \rightarrow 2NH_3$$

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	Standard molar entropy / J K^{-1} mol^{-1}
H_2	130.6
N_2	191.6
NH_3	192.3

☒ **A** –198.8

☒ **B** –129.9

☒ **C** +129.9

☒ **D** +198.8

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(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these has the **highest** standard molar entropy at 298 K and 1 atm pressure?

☒ **A** carbon dioxide, CO_2

☒ **B** copper, Cu

☒ **C** ethanol, C_2H_5OH

☒ **D** hydrogen, H_2

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The standard molar entropy, $S^{\ominus}$, of a substance

☒ **A** is not affected by changes of state or changes in temperature

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☒ **B** only changes when the temperature changes
☒ **C** only changes when the substance changes state
☒ **D** changes when the temperature changes and when the substance changes state

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which series does the standard molar entropy of the compounds increase?

☒ **A** $CaO(s) < H_2O(l) < CO_2(g) < SO_2(g)$
☒ **B** $CaO(s) < H_2O(l) < SO_2(g) < CO_2(g)$
☒ **C** $SO_2(g) < CO_2(g) < H_2O(l) < CaO(s)$
☒ **D** $CO_2(g) < SO_2(g) < H_2O(l) < CaO(s)$

(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence shows the elements in order of **increasing** standard entropy?

☒ **A** $He(g) < O_2(g) < Hg(l) < Os(s)$

☒ **B** $O_2(g) < He(g) < Hg(l) < Os(s)$

☒ **C** $Os(s) < Hg(l) < He(g) < O_2(g)$

☒ **D** $Os(s) < Hg(l) < O_2(g) < He(g)$

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When hydrogen and oxygen are mixed in a flask, there is no reaction.
When a platinum gauze is added to the flask, the mixture explodes.

It can be deduced that, in the absence of a platinum gauze, the mixture of hydrogen and oxygen is

- ☐ **A** kinetically stable and thermodynamically stable
- ☐ **B** kinetically stable and thermodynamically unstable
- ☐ **C** kinetically unstable and thermodynamically stable
- ☐ **D** kinetically unstable and thermodynamically unstable

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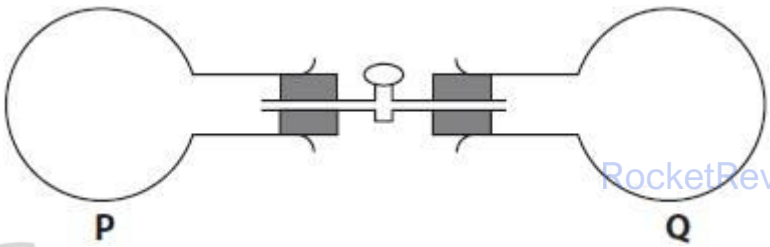
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Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two flasks, **P** and **Q**, are connected by a tube fitted with a tap.



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Flask **P** contains argon gas and flask **Q** is in vacuum.
Assume argon behaves as an ideal gas and therefore has no intermolecular forces between its atoms.

How do the entropy and energy of the system change when the tap is opened and the argon fills both flasks?

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	Entropy of the system	Energy of the system
☒ A	increases	decreases
☒ B	decreases	decreases
☒ C	increases	unchanged
☒ D	decreases	unchanged

(Total for question = 1 mark)

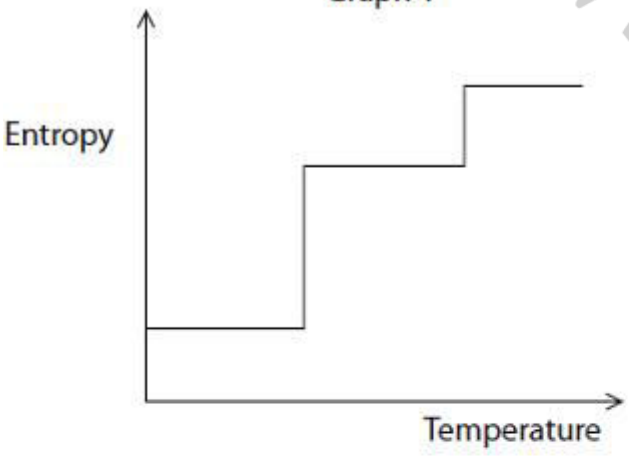
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Q15.

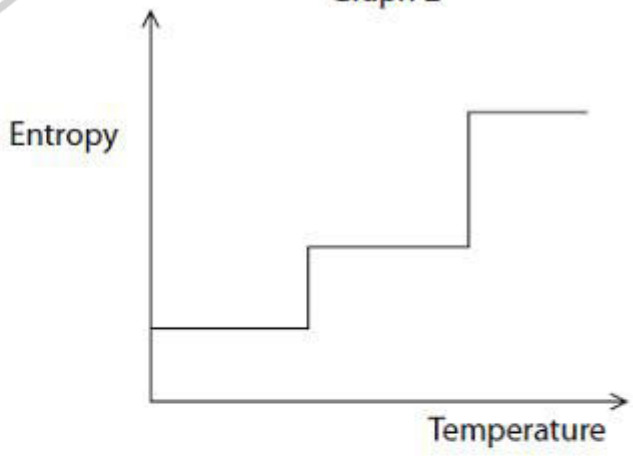
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which graph shows the entropy of a substance as it changes state from solid to liquid to gas?

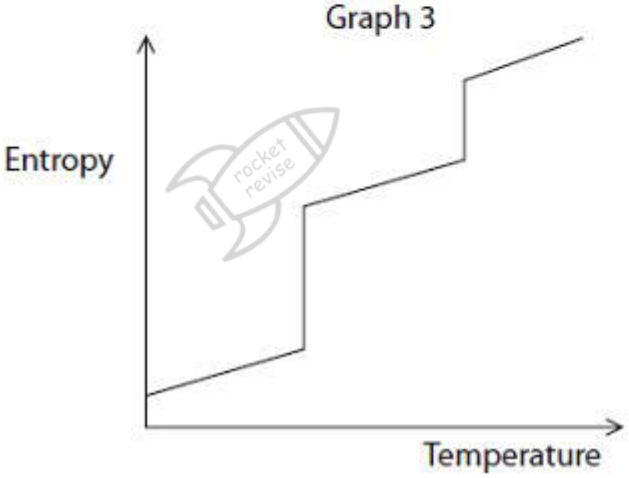
Graph 1



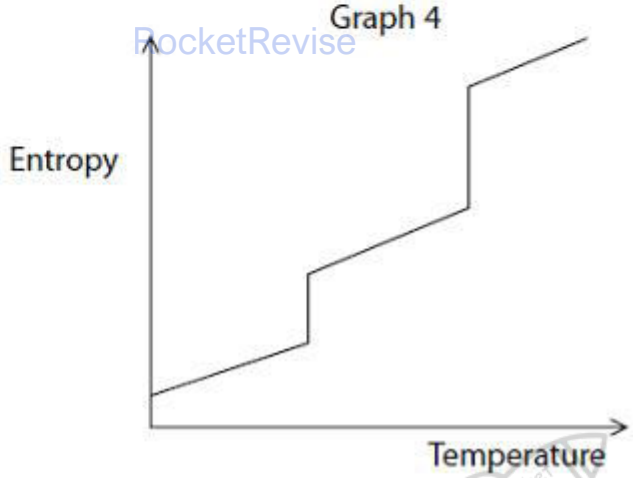
Graph 2



Graph 3



Graph 4



☒ **A** Graph 1
☒ **B** Graph 2

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☒ **C** Graph 3
☒ **D** Graph 4

(Total for question = 1 mark)

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Q16.

Nitrous oxide, N_2O, decomposes at high temperature to form nitrogen and oxygen.

$$N_2O(g) \rightleftharpoons N_2(g) + \tfrac{1}{2}O_2(g)$$

(a) (i) Some standard molecular entropy data are shown.

Substance	Standard molecular entropy $S^{\ominus}$ / $JK^{-1}mol^{-1}$
nitrogen, N_2	192
oxygen, O_2	205
nitrous oxide, N_2O	220

Calculate the standard entropy change of the system for the decomposition shown.
Include a sign and units in your answer.

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(2)

(ii) The standard enthalpy change of the forward reaction is −82 kJ mol^{-1}.
Calculate the entropy change of the surroundings at 2048 K.
Include a sign and units in your answer.

(2)

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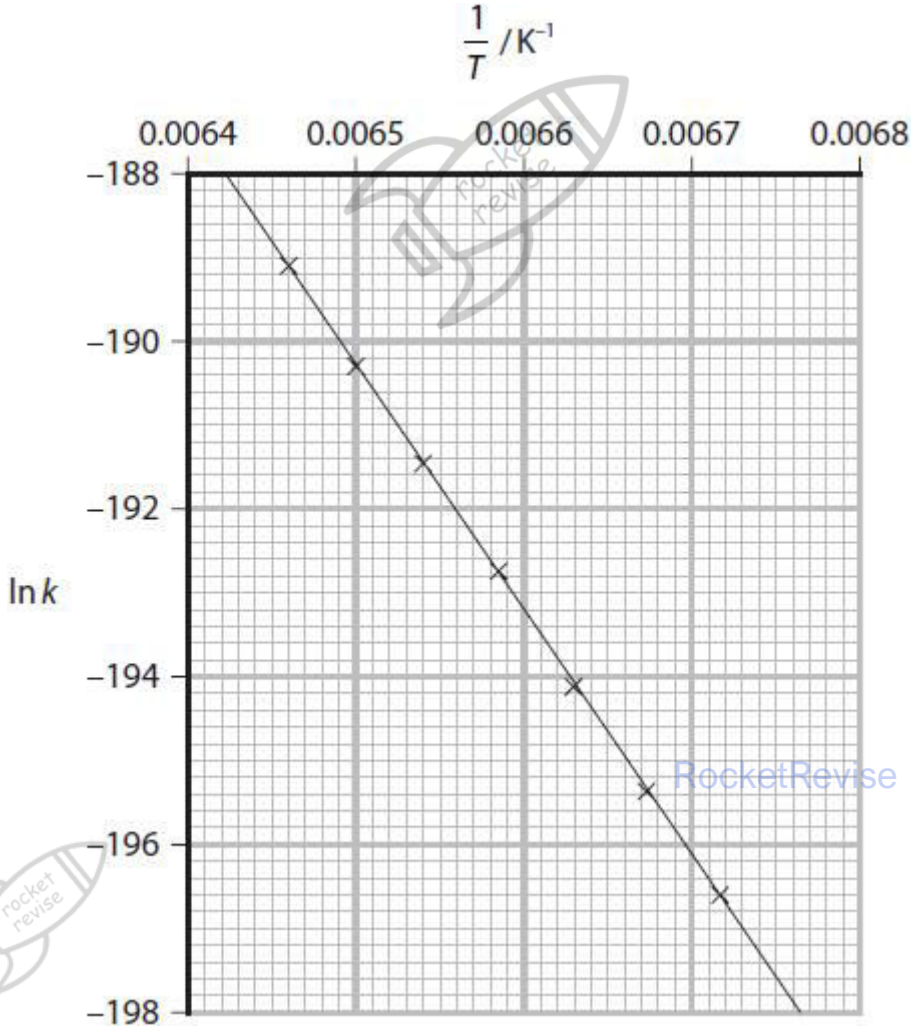
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(iii) Calculate the total entropy change of the reaction at 2048 K.
Include a sign and units in your answer.

(1)

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(b) Rate experiments on the decomposition of nitrous oxide produced the following graph.



Calculate the activation energy for the reaction in kJ mol^{-1}.
Include the value of the gradient.

$$\ln k = -\frac{E_a}{R}\frac{1}{T} + \text{constant}$$

$R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$

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(2)

(c) Explain whether or not this reaction occurs at 2048 K by considering the values calculated in (a) and (b).

(2)

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(Total for question = 9 marks)

Q17.

The mechanism for the rapid decomposition of hydrogen peroxide, H_2O_2, in the presence of iodide ions, is:

$$H_2O_2 + I^- \rightarrow IO^- + H_2O \qquad \text{equation 1}$$

$$H_2O_2 + IO^- \rightarrow I^- + H_2O + O_2 \qquad \text{equation 2}$$

(a) Write the overall equation for the reaction and hence state the role of the iodide ions.

(2)

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(b) In further experiments, a student calculated the rate constant for the decomposition of hydrogen peroxide at two different temperatures.

Temperature/ °C	Rate constant (k) / $dm^3\ mol^{-1}\ s^{-1}$
22.0	4.90×10^{-4}
47.0	2.92×10^{-3}

The rate constant (k) is related to the temperature, T, (in Kelvin) by the following equation:

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$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant} \qquad R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$$

Use the data in the table to calculate the activation energy, E_a, for the reaction by a non-graphical method.
Give your answer in kJ mol^{-1} and to an appropriate number of significant figures.
Include a sign in your answer.

(4)

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(c) If acid conditions are used, the decomposition of hydrogen peroxide proceeds by a different mechanism. The equation for this reaction is:

$$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$$

This reaction is first order with respect to both iodide ions and hydrogen peroxide.
The progress of this reaction is usually followed by adding a fixed quantity of sodium thiosulfate solution and a little starch solution to the reaction mixture, then timing the appearance of a blue-black colour. This is known as a clock reaction.

(i) Explain the formation of the blue-black colour and why its appearance is delayed.

(3)

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(ii) Although the oxidation of thiosulfate ions ($S_2O_3^{2-}$) by hydrogen peroxide is thermodynamically favourable, it does not take place in this clock reaction.
Suggest a reason for this.

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(1)

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(Total for question = 10 marks)

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Q18.

Which substance has the highest standard molar entropy?

(1)

- ☒ **A** ethane(g)
- ☒ **B** water(s)
- ☒ **C** water(l)
- ☒ **D** water(g)

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(Total for question = 1 mark)

Q19.

The element sulfur can exist in two solid, interchangeable, structural forms known as rhombic sulfur and monoclinic sulfur.

$$S(s, \text{rhombic}) \rightleftharpoons S(s, \text{monoclinic}) \qquad \Delta S_{total} = -0.307\ JK^{-1}mol^{-1}$$

The value of ΔS_{total} is for the forward reaction. What can be concluded from this information?

(1)

- ☒ **A** monoclinic sulfur will change quickly into rhombic sulfur
- ☒ **B** rhombic sulfur could change into monoclinic sulfur but nothing can be deduced about the rate
- ☒ **C** there can be no change of structural form as they are both solids
- ☒ **D** monoclinic sulfur could change into rhombic sulfur but nothing can be deduced about the rate

(Total for question = 1 mark)

Q20.

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This question is about some compounds of silicon.

(a) Silicon dioxide and magnesium react when heated strongly.

Reaction **1** $SiO_2(s) + 4Mg(s) \rightarrow Mg_2Si(s) + 2MgO(s)$

(i) Complete the table, indicating the type of bonding in the reactants and products of this reaction.

(2)

Substance	SiO_2	Mg	Mg_2Si	MgO
Bonding type			covalent	

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(ii) The entropy change of the system, ΔS_{system}, for Reaction **1** is −43.8 J K^{-1} mol^{-1}.
Suggest, with reference to the equation, why ΔS_{system} for this reaction is negative.

(2)

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(iii) The enthalpy change, ΔH, for Reaction **1** is −370 kJ mol^{-1}.
Calculate the entropy change of the surroundings, $\Delta S_{surroundings}$, in J K^{-1} mol^{-1}, for Reaction **1** at 23.0 °C.

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(2)

(iv) Calculate the total entropy change, ΔS_{total}, for Reaction **1** at 23.0 °C.
Give your answer to an appropriate number of significant figures, and in units of J K^{-1} mol^{-1}.

(2)

(v) Reaction **1** does not occur at room temperature due to its very high activation energy.
Suggest why the activation energy for Reaction **1** is very high.

(1)

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(b) Magnesium silicide, Mg_2Si, reacts with hydrochloric acid forming silane, SiH_4, and magnesium chloride.

(i) Write an equation for this reaction.
State symbols are **not** required.

(1)

(ii) Silane has a molecular structure.
Complete the table, giving the shape of a molecule of SiH_4 and its bond angle.

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(2)

Name of shape	
Bond angle	

(c) Silane spontaneously combusts in air at room temperature.

Reaction **2** $SiH_4(g) + 2O_2(g) \rightarrow SiO_2(s) + 2H_2O(l)$
Entropy data for this reaction are shown.

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Species	$SiH_4(g)$	$O_2(g)$	$SiO_2(s)$	$H_2O(l)$
$S^\ominus$ / $JK^{-1}mol^{-1}$	204.5	205.0	41.8	69.9

(i) Calculate the entropy change of the system, ΔS_{system}, in J K^{-1} mol^{-1}, for Reaction **2**.

(2)

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(ii) State why the entropy change of the surroundings for Reaction **2** is highly positive, in terms of the bond strengths of the reactants and products.

(1)

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(Total for question = 15 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equations for three reactions involving hydrogen are shown.

J $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

K $N_2(g) + 2H_2(g) \rightarrow N_2H_4(l)$

L $I_2(s) + H_2(g) \rightarrow 2HI(g)$

What is the order of **increasing** standard entropy change of the system, $\Delta S^\ominus_{system}$, for these reactions?

- ☒ **A** J, K, L
- ☒ **B** K, L, J
- ☒ **C** K, J, L
- ☒ **D** L, K, J

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(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What are the signs of the entropy changes at 373 K when water vapour condenses?

$$H_2O(g) \rightarrow H_2O(l)$$

		ΔS_{system}	$\Delta S_{surroundings}$
☐	**A**	positive	positive
☐	**B**	positive	negative
☐	**C**	negative	positive
☐	**D**	negative	negative

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(Total for question = 1 mark)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Methane reacts with steam to produce carbon dioxide and hydrogen.

$$CH_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4H_2(g)$$

The standard molar entropies of the reactants and products are given in the table.

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Substance	$S^{\ominus}$ / $JK^{-1}mol^{-1}$
$CH_4(g)$	186
$H_2O(g)$	189
$CO_2(g)$	214
$H_2(g)$	131

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The value of $\Delta S^{\ominus}_{system}$ for this reaction, in J K^{-1} mol^{-1}, is

- ☒ **A** −174
- ☒ **B** −30
- ☒ **C** +30
- ☒ **D** +174

(Total for question = 1 mark)

Q24.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Some thermodynamic information is shown for four reactions, W, X, Y and Z, at 298 K.

Reaction	ΔH	$T\Delta S_{system}$
W	negative and with a larger negative value than $T\Delta S_{system}$	negative
X	negative	positive
Y	positive	positive and with a smaller positive value than ΔH
Z	positive	negative

Which of these reactions are feasible at 298 K?

- ☒ **A** X only
- ☒ **B** W and X only
- ☒ **C** Y only
- ☒ **D** W and Z only

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(Total for question = 1 mark)

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pair of values will **always** give a thermodynamically feasible reaction?

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	Entropy change of the system	Enthalpy change of reaction
☐ **A**	negative	positive
☐ **B**	negative	negative
☐ **C**	positive	negative
☐ **D**	positive	positive

(Total for question = 1 mark)

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Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these substances would be expected to have the greatest standard molar entropy?

- ☐ **A** FeO
- ☐ **B** FeS
- ☐ **C** $FeCl_2$
- ☐ **D** $FeCl_3$

(Total for question = 1 mark)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which has the greatest molar entropy?

- ☐ **A** Hg(l)
- ☐ **B** $H_2O(l)$
- ☐ **C** NO(g)
- ☐ **D** $NO_2(g)$

(Total for question = 1 mark)

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